KIRAN DAMODARAN

AI Engineer | Computer Vision | LLM, RAG & Agents | Robotics & Edge AI (ROS2, Jetson) | Python

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SUMMARY

AI Engineer with 3+ years across robotics, computer vision, multimodal LLM applications, RAG, and multi-agent systems. Builds the perception, speech, and ML pipelines behind a customer-facing service robot: document and ID scanning at 97% accuracy in live use, a multilingual voice pipeline under 2 seconds, and a ROS2 AI stack on NVIDIA Jetson. Also built FieldForge, a public suite of RAG, agent, MCP/A2A, and LLMOps services.

EXPERIENCE

AI Engineer | FWM Information Technology LLC *01/2026 - Present | Dubai, UAE*

- Built a 4-mode document/ID scan pipeline (AUTO, OCR, ID, ONESHOT) on the Robin service robot using Tesseract OCR, a vision LLM API, and OpenCV, reaching 97% accuracy in live customer-facing use.
- Designed a multilingual conversation pipeline (Vosk speech-to-text, LLM API intent routing, Piper text-to-speech) for English, Arabic, and Farsi, keeping the full response cycle under 2 seconds.
- Packaged the ROS2 AI stack as 8 independently restartable lifecycle nodes on NVIDIA Jetson, improving responsiveness 55% and enabling production hot restarts with no robot downtime.

Research Scientist | Colorado School of Mines, Payne Institute for Public Policy *06/2024 - 06/2025 | Colorado, USA*

- Built Isolation Forest and LLM-based root-cause pipelines that cut anomaly-to-insight time 30%.
- Cut methane-emissions analytics turnaround 60% at the METEC oil and gas facility with a Python dead-letter-queue pipeline and a spike-detection algorithm (99% accuracy, 50+ sensors).

Graduate Research Assistant | Colorado State University (METEC) *01/2023 - 05/2024 | Colorado, USA*

- Improved network uptime 45% via QGIS-based telemetry diagnostics across 50+ embedded systems and cut security vulnerabilities 40% designing an Attribute-Based Access Control framework for DER, PLC, and SCADA systems.

Associate Data Analyst, AI & Analytics | Cognizant Technology Solutions (Client: RBC) *06/2021 - 07/2022 | Chennai, India*

- Lifted batch-processing throughput 25% with IBM InfoSphere DataStage ETL pipelines for RBC's enterprise data warehouse; automated daily transaction validation in JCL and built Informatica CI/CD pipelines that cut deployment errors 20%.

KEY PROJECTS | INDEPENDENT FULL-STACK BUILDS

FieldForge AI Suite: RAG, agents, MCP/A2A & LLMOps (public on GitHub) *(FastAPI, Qdrant, Ollama, MCP, pytest)*

- Built five connected AI services in Python: grounded Q&A (BM25 plus Qdrant/Ollama hybrid retrieval, Recall@5 1.0, 13/13 adversarial guardrail cases), an incident-investigation agent, a two-agent A2A/MCP mesh with a real MCP server, and an LLMOps gate that blocks deployment on failing evals; 159 tests in CI.

AI Invoice-Review Copilot: multi-agent RAG pipeline, security & testing *(FastAPI, PostgreSQL, Docker, Playwright)*

- Built a multi-agent invoice-review copilot: hybrid retrieval grounds LLM explanations in contract clauses, an evidence verifier rejects unsupported findings, one provider layer serves Anthropic, Azure OpenAI, Ollama; 130/130 tests in CI on a live LLM.

Karma Restaurant HR *(ASP.NET Core, EF Core, SQL Server, ASP.NET Core Identity)*

- Built and shipped a live workforce platform for a 30-person restaurant team running real payroll: Face ID attendance with liveness detection, GPS-geofenced check-ins, a 5-stage payroll approval chain, and a UAE EOSB calculator; 42 xUnit tests including IDOR and payroll-lock adversarial scenarios.

ZS School *(ASP.NET Core (.NET 9), EF Core, SQL Server)*

- Built a school management platform (4 role portals, analytics dashboards, Excel export, Android companion app) with a versioned IIS release pipeline, health checks, and a rollback runbook.

Websites & software *(WordPress, Django, PostgreSQL)*

- Run WordPress sites for Zenon Petroleum (zenonpetroleum.ae), Zenon Lubricants (zenonlub.com) and Karma Dubai (karma-dubai.com); built RentFlow, a Django/PostgreSQL rental ledger platform.

EDUCATION

M.S. in Computer Engineering | Colorado State University | GPA 3.71 *08/2022 - 05/2024 | USA*

B.Tech in Electronics and Communication Engineering (ECE) | SRM Institute of Science and Technology *2017 - 2021 | Chennai, India*

TECHNICAL SKILLS

- **Programming & AI/ML:** Python, C#, SQL, R, JavaScript/TypeScript, Dart, Machine Learning, PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO (YOLOv8), Tesseract OCR, Isolation Forest, Computer Vision, Time-Series Analysis, Forecasting, Anomaly Detection
- **LLM, RAG & Agents:** LLM APIs, RAG, Hybrid Retrieval (BM25, vector), Qdrant (vector database), Embeddings, MCP, A2A, Multi-Agent Orchestration, AI Agents, Prompt Engineering, LLM Evaluation, LLMOps, Guardrails, Prompt-Injection Defense, Hallucination Guard, Vision LLM, Anthropic, Azure OpenAI, Ollama
- **Web & Mobile:** ASP.NET Core MVC, Entity Framework Core, ASP.NET Core Identity, FastAPI, Django, WordPress, Flutter (iOS/Android), REST APIs, WebSockets, SQLite
- **Robotics & Edge:** ROS2 (Humble), NVIDIA Jetson (Orin NX), TensorRT, Embedded Systems, Autonomous Systems
- **Data, DevOps & Security:** Docker, Git, GitHub Actions CI/CD, AWS, Linux, IIS, PostgreSQL, SQL Server, Row-Level Security, RBAC, Prometheus, Trivy, gitleaks, pip-audit, Playwright, xUnit, pytest, Snowflake, DataStage, Informatica, JCL, ETL/ELT, pandas